Scientific AI

D-Lab AI Pulse Workshop

The Promise and the Problem

There has been a lot of interest in using AI tools for academic research.

What they're great at:

- Brainstorming and drafting
- Explaining complex concepts
- Writing and editing code
- Summarizing long documents

But for academic research, there's a critical flaw:

These tools can present fabricated information as fact and without warning.

Today: Why this happens, and tools designed specifically to avoid it.

Today's Roadmap

1. **The Problem**
   - Why general AI hallucinates citations
   - Real-world consequences
2. **The Solution**
   - How specialized research tools work differently
   - Perplexity, Consensus, Elicit
3. **The Frontier**
   - Kosmos: Autonomous research agents
   - What's coming next

Live demos throughout.

The Hallucination Problem

Large Language Models predict plausible text, not accurate text.

They are not designed to verify whether their output is true, and lack uncertainty quantification.

Why it's convincing: A fake citation like "Smith & Johnson (2020)" follows patterns from millions of real papers — common surnames, recent year, proper format. The model generates *plausible* text, not *verified* text.

The numbers:

- ChatGPT fabricates **18–55%** of citations depending on model version[1]
- **24–43%** of real references contain substantive errors[1]
- Specialized legal AI tools hallucinate in **17–33%** of queries[2]

[1]Walters & Wilder (2023), *Scientific Reports*; [2]Magesh et al. (2025), *J. Empirical Legal Studies*

Real-World Consequences

In consulting:

- Deloitte Australia (2025): Welfare report with fabricated references
- Deloitte Canada (2025): Healthcare report with fake citations

In law:

- Mata v. Avianca (2023): 6 nonexistent cases cited
- California appeal (2025): 21 of 23 quoted cases fabricated
- Hundreds of errant legal citations flagged in US courts to date

In academia:

- 100+ hallucinated citations found across 51 accepted NeurIPS 2025 papers
- Fake citations get published, cited by others, and the problem cascades

Why Specialized Tools?

General AI (ChatGPT, Claude, Gemini)	**Academic AI** (Elicit, Consensus, Perplexity)
Generates plausible text	Retrieves from verified sources
No database verification	200M+ papers (Semantic Scholar, OpenAlex, PubMed)
Citations may be fabricated	Citations are real and verifiable

Key difference: *Retrieval* from verified sources vs. *generation* from patterns.

Three Tools, One Workflow

All three: Ask a question → get papers with AI summaries → filter → export citations

All three: Semantic search, sentence-level citations, citation export

Tool	Best For	Unique Feature
Perplexity	Quick context, broad questions	Web + academic search
Consensus	"What does evidence say?"	Consensus Meter
Elicit	Systematic reviews	Structured data extraction

Limitations: Smaller coverage than Google Scholar or library databases. Better for established topics. Supplement, don't replace, traditional database searches.

Perplexity

Broad search: Academic papers + web sources + news

Features:

- **Model selection:** Choose between Claude, GPT, Grok — different strengths per query
- **Academic Mode:** Restricts sources to peer-reviewed literature
- **Deep Research:** Breaks question into sub-questions, synthesizes across dozens of sources
- **Model Council:** Queries multiple models and synthesizes their answers

Best for: Background research, exploring unfamiliar topics, quick fact-checking

Demo:

```
What are the main criticisms of DSGE models
in macroeconomics?
```

Perplexity Spaces

Persistent, shareable research environments.

What you can do:

- **Upload files** — syllabi, readings, datasets, codebooks, notes
- **Invite collaborators** — classmates, lab members, advisors
- **Ask questions** grounded in your uploaded materials + the full web
- **Build a shared research trail** everyone can see and build on

Example workflow:

Create a Space for your final project → upload your readings and dataset → invite your partner → now you both ask questions and build a shared evidence base.

Consensus

Focused on academic literature: Peer-reviewed papers, preprints, and conference proceedings

Includes: Wiley, Taylor & Francis, and other major publishers

Features: Consensus Meter, import from Zotero, upload your own papers

Premium available for Berkeley

Best for: Evidence-based questions, checking scientific consensus, policy debates

Demo:

```
Does minimum wage reduce employment?
```

Elicit

Designed for systematic reviews: Screening, data extraction, report generation

Workflows:

- **Extract Data** — build custom tables from papers (sample sizes, effect sizes, methods)
- **Competitive Landscape** — compare products or approaches across academic and non-academic sources
- **General Research Agent** — flexible agent that breaks down any research question

Demo:

```
What methods are used to measure inflation
expectations?
```

JSTOR AI Research Tool

AI built on top of a library database you already have access to.

What it does:

- Assess content relevance — surfaces key points and arguments from texts
- Discover related topics and materials across the JSTOR corpus
- Conversational — use natural language to search or ask questions about what you're reading

Demo:

```
What is the relationship between central bank
independence and inflation?
```

When to Use Something Else

These tools are slower than ChatGPT. That's by design — more verification takes more time. Use ChatGPT/Claude for brainstorming and drafting; use these tools for sourcing.

Coverage gaps:

- Very recent papers (<3 months) — indexing takes time
- Niche topics with thin literature
- Non-English publications

Use Google Scholar or your library database when you need:

- Exhaustive coverage — dissertations, theses, grey literature, working papers
- Known-item searches — you already have a title or DOI
- Full-text access through Berkeley's subscriptions
- Non-English or region-specific literature

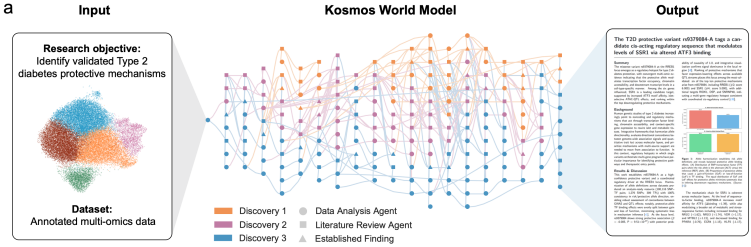
Best approach: Start broad with these tools, verify and fill gaps with traditional databases.

Discussion

1. **Teach or restrict?** Students and researchers will use these tools regardless. Are we better off trying to stop them, or teaching them to use them well?
2. **Industry is pushing AI.** Not just welcoming it, but actively requiring it. How does that change what and how we teach?
3. **The next shift.** Information went from low volume, high signal-to-noise, to high volume, low signal-to-noise, to on-demand volume and unobservable signal-to-noise ratio. We can't imagine research without Google today. What will research look like in 25 years?
4. **Equity and access.** Free tiers exist, but premium features cost money. Does AI narrow the gap between well-funded and under-funded researchers, or widen it?

What is Kosmos?

An autonomous research agent from FutureHouse / Edison Scientific



Two modes:

- **Kosmos** (full automation): Submit a question + data (up to 15GB), receive a report in ∼12 hours
- **Interactive agents:** Literature synthesis, data analysis, molecular design, precedent search

Access: Free for academics with .edu email (650 credits/month)

Kosmos: Discoveries

Mitchener et al. (2025), arXiv:2511.02824 — 7 discoveries across 4 fields. Estimated equivalent of ∼6 months of human research per run.

Agent	Lines of Code	Papers Read
Kosmos	42,500 ±7,280	1,500 ±1,120
Robin	4,310 ±344	1,530 ±289
Finch	301 ±36	0 ±0
PaperQA2	0 ±0	33 ±29

Novel findings:

- SOD2 causally reduces myocardial fibrosis (Mendelian randomization)
- GWAS variant regulates SSR1 in Type 2 diabetes (multi-omic analysis)
- Temporal ordering of ECM decline during tau accumulation in Alzheimer's
- Flippase downregulation triggers neuron loss in entorhinal cortex

Independent reproductions:

- Nucleotide metabolism in hypothermic brain protection
- Humidity as critical factor in perovskite solar cell fabrication
- Log-normal connectivity distributions across 5 species

Kosmos: Best Practices

1. **Clear, feasible research objective.** Single, well-defined goal with scope for iterative testing. Should require weeks or months of human work, not a single analysis.
2. **Provide enough context.** Phrase it as you would for an experienced colleague who just joined your team. Include field-specific nuances and experimental design choices.
3. **Describe your data.** Label columns intuitively or provide a codebook. A collaborator should be able to interpret the data without further clarification.
4. **Complex data works best.** High-dimensional data (scRNA-seq, proteomics, multi-sample environmental data). Multiple datasets welcome, up to 15GB total.
5. **Processed data.** Kosmos can do QC, but works best on cleaned, processed data. Not recommended for raw sequencing or raw imaging.
6. **Iterate.** Start with a familiar dataset. Runtime scales with data size, so begin small.

Full guidelines: `https://edisonscientific.com/articles/kosmos-best-practices-guidelines`

Resources: Tools

Tools covered today:

- Perplexity: `https://perplexity.ai/`
- Consensus: `https://consensus.app/`
- Elicit: `https://elicit.com/`
- JSTOR AI: `https://www.jstor.org/` (Free via Berkeley Library)
- Kosmos / Edison Scientific: `https://edisonscientific.com/`

Also mentioned:

- Research Rabbit: `https://www.researchrabbit.ai/`
- Connected Papers: `https://www.connectedpapers.com/`
- Litmaps: `https://www.litmaps.com/`
- Scite.ai: `https://scite.ai/`
- SciSpace: `https://typeset.io/`

Resources: References

Papers and data cited:

- Walters & Wilder (2023). "Fabrication and Errors in the Bibliographic Citations Generated by ChatGPT." *Scientific Reports*. https://doi.org/10.1038/s41598-023-41032-5
- Magesh et al. (2025). "Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools." *J. Empirical Legal Studies*.
- Ansari (2026). "Compound Deception in Elite Peer Review: A Failure Mode Taxonomy of 100 Fabricated Citations at NeurIPS 2025." arXiv:2602.05930. https://arxiv.org/abs/2602.05930
- Mitchener et al. (2025). "Kosmos: Autonomous Research Agent." arXiv:2511.02824.
- Charlotin, D. AI Hallucination Cases Database (912 cases). https://www.damiencharlotin.com/hallucinations/

D-Lab: https://dlab.berkeley.edu/